

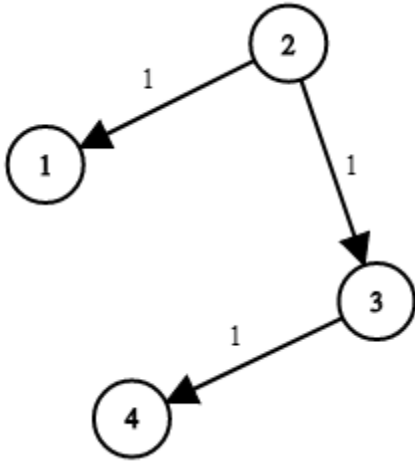
743. Network Delay Time

Solved Medium

You are given a network of n nodes, labeled from 1 to n . You are also given times, a list of travel times as directed edges $\text{times}[i] = (u_i, v_i, w_i)$, where u_i is the source node, v_i is the target node, and w_i is the time it takes for a signal to travel from source to target.

We will send a signal from a given node k . Return the *minimum* time it takes for all the n nodes to receive the signal. If it is impossible for all the n nodes to receive the signal, return -1.

Example 1:



Input: $\text{times} = [[2,1,1],[2,3,1],[3,4,1]]$, $n = 4$, $k = 2$

Output: 2

Example 2:

Input: $\text{times} = [[1,2,1]]$, $n = 2$, $k = 1$

Output: 1

Example 3:

Input: $\text{times} = [[1,2,1]]$, $n = 2$, $k = 2$

Output: -1